Virtual Reality (VR)

Summary

Virtual Reality refers to technologies that allow users to experience and interact with a computer-created world. Unlike normal screens, VR surrounds your senses especially sight and sound to trick your brain into feeling like you're inside a virtual space. It blends visual, audio, and motion tracking techniques to simulate real or imaginary environments, making the experience immersive and realistic. The core characteristics include:

- Immersion: surrounding the user visually & audibly
- Interactivity: Ability to move, explore, and manipulate virtual objects.
- Real-time rendering: Environment updates instantly based on user movement.
- Presence: Feeling "inside" the virtual world

Working of VR

1. Display & optics:
   → High resolution screens
   → Wide field of view (FOV)
   → Lenses that shape images for a 3D effect

2. Motion Tracking
   → Tracks head, hand, and body movement
   → Uses gyroscopes, accelerometers and external sensors
   → Ensures the virtual world reacts to the user in real tim

3. Input devices
- Motion controllers • Haptic gloves • Eye-tracking systems
- Treadmills or body trackers

4. Software & Engines
- Unity, Unreal Engine for 3D rendering
- Algorithms for environment physics and motion response
- AI for realistic interactions.

The various types of VR are
→ Immersive VR system
→ Semi - immersive VR system
→ Non - immersive VR system

Advantages of VR
- Highly immersive learning and training
- safe environment for risky simulations
- Enhances creativity and innovation
- Makes remote collaboration effective

Limitations of VR
- High cost of equipment
- Motion sickness in some users
- Requires powerful hardware for high-end VR
- Limited physical movement space
- Long usage may cause eye strain

Future of VR
The future of VR is promising with advancements in:
- Lightweight, wireless VR headsets
- 8K displays and improved field of view
- Advanced haptic suits for full-body sensation
- AI - driven realistic environments